

File generation

LESSON 08 · MODULE 2 — MULTIMODAL

What you'll learn

- How `file_output=True` switches an agent from text/image to document mode
- Why the file *format* is a property of the prompt, not the agent
- How the same artifact retrieval pattern from image generation reuses for documents
- Where document generation is most reliable — and where the model is most likely to drift

Key concepts

Same agent, different artifact. Add `file_output=True` to `create_agent` and the same `agent.run()` returns a document. PDF, CSV, DOCX, XLSX — whatever you ask for in the prompt.

Format lives in the prompt, not the agent. The agent doesn't know what format it's producing until you tell it in the message. "Format as a CSV" → CSV. "Format as a PDF" → PDF. Same agent can generate one of each across two calls without recreating it.

Artifact retrieval is identical to image generation.

`response.has_files` to check, iterate `response.files`, call `artifact.download(path)` to write to disk. Each artifact has `format_type` so you can branch on document vs image vs audio.

Tabular formats are most reliable. Five columns and five rows? The model has very little room to drift. The schema is rigid; cells get populated cleanly. PDFs give the model more room to embellish — it will invent metrics, headings, "outlook" sections you didn't ask for. Useful when you want a polished one-pager, dangerous when you need accuracy.

Numbers are AI-generated. If you ask for "revenue highlights," the dollar figures will be plausible but invented. Don't ship them as truth.

Code shown in this lesson

PYTHON

TAKEAWAY

Same agent, different artifact.

TAKEAWAY

Format lives in the prompt, not the agent.

TAKEAWAY

Artifact retrieval is identical to image generation.

TAKEAWAY

Tabular formats are most reliable.

TAKEAWAY

Numbers are AI-generated.

```
# Agent in document mode
agent = studio.create_agent(
    name="document-generator",
    provider="gpt-4o",
    role="business writer",
    goal="produce structured business documents",
    instructions="Be concise and well-formatted.",
    file_output=True,
)
```

PYTHON

```
# PDF – format clause is in the prompt
response = agent.run("""
    Write a Q3 sales QBR with revenue highlights, top three wins, and one risk.
    Format as a PDF.
    """)

for artifact in response.files:
    artifact.download(safe_file_name(artifact.name))
```

PYTHON

```
# CSV – same agent, different format clause
response = agent.run("""
    Generate five rows of project status data with columns:
    project, owner, status, priority, due_date.
    Format as a CSV.
    """)
```

TRY THIS

- 1 Same agent, same prompt content — *"five rows of project status data with columns: project, owner, status, priority, due_date"* — but ask once for CSV and once for DOCX. Compare. Note where the model embellishes more (you should see CSV stay rigid and DOCX add formatting flourish).
- 2 Ask the doc agent for a 2-page PDF report on a topic you actually know well (your last sprint, your favourite product). Read the output critically — circle every metric the model invented vs every fact that was grounded.